

Translated

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Catalan Numbers

Catalan numbers is a number sequence, which is found useful in a number of combinatorial problems, often involving recursively-defined objects.

This sequence was named after the Belgian mathematician [Catalan](#), who lived in the 19th century. (In fact it was known before to Euler, who lived a century before Catalan).

The first few Catalan numbers C_n (starting from zero):

1, 1, 2, 5, 14, 42, 132, 429, 1430, $\dots$

Application in some combinatorial problems

The Catalan number C_n is the solution for

- Number of correct bracket sequence consisting of n opening and n closing brackets.
- The number of rooted full binary trees with $n + 1$ leaves (vertices are not numbered). A rooted binary tree is full if every vertex has either two children or no children.
- The number of ways to completely parenthesize $n + 1$ factors.
- The number of triangulations of a convex polygon with $n + 2$ sides (i.e. the number of partitions of polygon into disjoint triangles by using the diagonals).
- The number of ways to connect the $2n$ points on a circle to form n disjoint chords.
- The number of [non-isomorphic](#) full binary trees with n internal nodes (i.e. nodes having at least one son).
- The number of monotonic lattice paths from point $(0, 0)$ to point (n, n) in a square lattice of size $n \times n$, which do not pass above the main diagonal (i.e. connecting $(0, 0)$ to (n, n)).
- Number of permutations of length n that can be [stack sorted](#) (i.e. it can be shown that the rearrangement is stack sorted if and only if there is no such index $i < j < k$, such that $a_k < a_i < a_j$).
- The number of [non-crossing partitions](#) of a set of n elements.
- The number of ways to cover the ladder $1 \dots n$ using n rectangles (The ladder consists of n columns, where i^{th} column has a height i).